

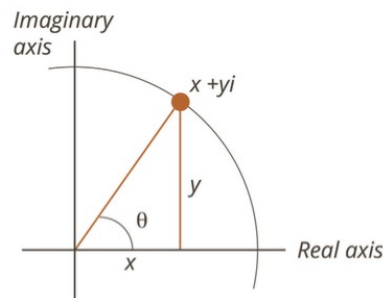
HW 6.9.2: Complex Numbers in Polar Form and Euler Numbers

Rewrite each complex number from polar form into $a + bi$ form.

1. $3e^{\frac{5\pi}{6}i}$

2. $3e^{\frac{\pi}{2}i}$

3. $6e^{\frac{\pi}{6}i}$



Euler's Formula

$$z = re^{i\theta} = r \cos(\theta) + ir \sin(\theta)$$

4. $7e^{\frac{4\pi}{3}i}$

5. $3e^{\frac{5\pi}{4}i}$

6. $6e^{\frac{5\pi}{4}i}$

Rewrite each complex number into polar $re^{i\theta}$ form.

7. 6

8. -2

9. $-4i$

10. $3i$

11. $2 + 2i$

12. $2\sqrt{2} + 2\sqrt{2}i$

13. $-3\sqrt{2} + 3\sqrt{2}i$

14. $-2 + 2\sqrt{3}i$

15. $5\sqrt{3} - 5i$

16. $-3 - 3\sqrt{3}i$

17. $2\sqrt{3} + 2i$

18. $-\sqrt{3} - i$

Selected Answers:

$$1. \quad 3e^{\frac{5\pi}{6}i} = 3\cos\left(\frac{5\pi}{6}\right) + i3\sin\left(\frac{5\pi}{6}\right) = -\frac{3\sqrt{3}}{2} + \frac{3}{2}i$$

$$3. \quad 6e^{\frac{\pi}{6}i} = 6\cos\left(\frac{\pi}{6}\right) + i6\sin\left(\frac{\pi}{6}\right) = 6\left(\frac{\sqrt{3}}{2}\right) + i6\left(\frac{1}{2}\right) = 3\sqrt{3} + 3i$$

$$5. \quad 3e^{\frac{5\pi}{4}i} = 3\cos\left(\frac{5\pi}{4}\right) + i3\sin\left(\frac{5\pi}{4}\right) = 3\left(-\frac{\sqrt{2}}{2}\right) + i3\left(-\frac{\sqrt{2}}{2}\right) = -\frac{3\sqrt{2}}{2} - \frac{3\sqrt{2}}{2}i$$

7. $6 = x + yi$ so $x = 6$ and $y = 0$. Also, $r^2 = x^2 + y^2 = 6^2 + 0^2 = 6^2$ so $r = 6$ (since $r \geq 0$). Also, using $x = r\cos(\theta)$: $6 = 6\cos(\theta)$, so $\theta = 0$. So the polar form is $6e^{0i}$.

9. $-4i = x + yi$ so $x = 0$ and $y = -4$. Also, $r^2 = x^2 + y^2 = 0^2 + (-4)^2 = 16$ so $r = 4$ (since $r \geq 0$).

Using $y = r\sin(\theta)$: $-4 = 4\sin(\theta)$, so $\sin(\theta) = -1$, so $\theta = \frac{3\pi}{2}$. So the polar form is $4e^{\frac{3\pi}{2}i}$.

11. $2 + 2i = x + yi$ so $x = y = 2$. Then $r^2 = x^2 + y^2 = 2^2 + 2^2 = 8$, so $r = 2\sqrt{2}$ (since $r \geq 0$). Also $x = r\cos(\theta)$ and $y = r\sin(\theta)$, so:

$$2 = 2\sqrt{2}\cos(\theta) \text{ so } \cos(\theta) = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}, \text{ and } 2 = 2\sqrt{2}\sin(\theta) \text{ so } \sin(\theta) = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$$

Therefore (x, y) is located in the quadrant I, or $\theta = \frac{\pi}{4}$. So the polar form is $2\sqrt{2}e^{\frac{\pi}{4}i}$.

$$13. \quad 6e^{\frac{3\pi}{4}i}$$

$$15. \quad 10e^{\frac{11\pi}{6}i}$$

$$17. \quad 4e^{\frac{\pi}{6}i}$$

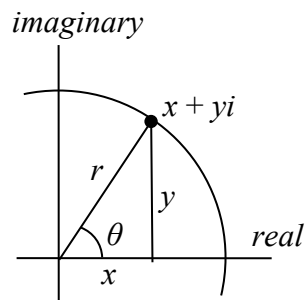
Notes:

Polar Form of Complex Numbers

Remember, because the complex plane is analogous to the Cartesian plane that we can think of a complex number $z = x + yi$ as analogous to the Cartesian point (x, y) and recall how we converted from (x, y) to polar (r, θ) coordinates in the last section.

Bringing in all of our old rules we remember the following:

$$\begin{aligned}\cos(\theta) &= \frac{x}{r} & x &= r \cos(\theta) \\ \sin(\theta) &= \frac{y}{r} & y &= r \sin(\theta) \\ \tan(\theta) &= \frac{y}{x} & x^2 + y^2 &= r^2\end{aligned}$$



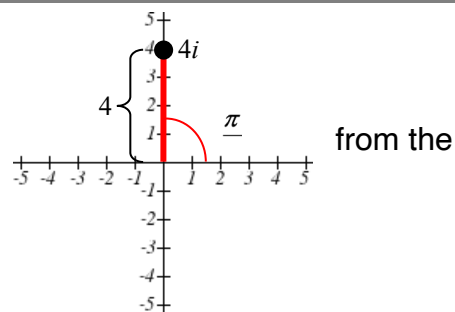
With this in mind, we can write $z = x + yi = r \cos(\theta) + ir \sin(\theta)$.

Example 6

Express the complex number $4i$ using polar coordinates.

On the complex plane, the number $4i$ is a distance of 4 from the origin at an angle of $\frac{\pi}{2}$, so $4i = 4 \cos\left(\frac{\pi}{2}\right) + i4 \sin\left(\frac{\pi}{2}\right)$

Note that the real part of this complex number is 0.



In the 18th century, Leonhard Euler demonstrated a relationship between exponential and trigonometric functions that allows the use of complex numbers to greatly simplify some trigonometric calculations. While the proof is beyond the scope of this class, you will likely see it in a later calculus class.

Polar Form of a Complex Number and Euler's Formula

The **polar form of a complex number** is $z = re^{i\theta}$, where **Euler's Formula** holds:

$$z = re^{i\theta} = r \cos(\theta) + ir \sin(\theta)$$

Similar to plotting a point in the polar coordinate system we need r and θ to find the polar form of a complex number.

Example 7

Find the polar form of the complex number -7 .

Treating this as a complex number, we can consider the unsimplified version $-7+0i$.

Plotted in the complex plane, the number -7 is on the negative horizontal axis, a distance of 7 from the origin at an angle of π from the positive horizontal axis.

The polar form of the number -7 is $7e^{i\pi}$.

Plugging $r = 7$ and $\theta = \pi$ back into Euler's formula, we have:

$$7e^{i\pi} = 7\cos(\pi) + 7i\sin(\pi) = -7 + 0i = -7 \text{ as desired.}$$

Example 8

Find the polar form of $-4 + 4i$.

On the complex plane, this complex number would correspond to the point $(-4, 4)$ on a Cartesian plane. We can find the distance r and angle θ as we did in the last section.

$$r^2 = x^2 + y^2$$

$$r^2 = (-4)^2 + 4^2$$

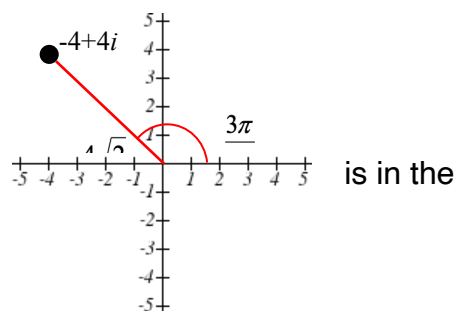
$$r = \sqrt{32} = 4\sqrt{2}$$

To find θ , we can use $\cos(\theta) = \frac{x}{r}$

$$\cos(\theta) = \frac{-4}{4\sqrt{2}} = -\frac{\sqrt{2}}{2}$$

This is one of known cosine values, and since the point is in the second quadrant, we can conclude that $\theta = \frac{3\pi}{4}$.

The polar form of this complex number is $4\sqrt{2}e^{\frac{3\pi}{4}i}$.



We can also use $\tan(\theta) = \frac{y}{x}$ to find the angle, but we must remember to check the quadrant.

Try it Now

3. Write $1 + \sqrt{2}i$ in polar form.

Example 9

Write $2e^{\frac{2\pi}{3}i}$ in complex $a + bi$ form.

$$2e^{\frac{2\pi}{3}i} = 2\cos\left(\frac{2\pi}{3}\right) + i2\sin\left(\frac{2\pi}{3}\right)$$

Evaluate the trig functions

$$= 2\left(-\frac{1}{2}\right) + i2\left(\frac{\sqrt{3}}{2}\right)$$

Simplify

$$= -1 + i\sqrt{3}$$

The polar form of a complex number provides a powerful way to compute powers and roots of complex numbers by using exponent rules from algebra. To compute a power of a complex number:

- 1) Convert to polar form
- 2) Raise to the power, using exponent rules to simplify
- 3) Convert back to $a + bi$ form, if needed